



MSc DSAI 20242025 AIEDA323 Biomedical Signal Processing Computer Labs (2 Adaptive noise canceling) - Professor Vicente Zarzoso

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1 2 Adaptive noise canceling

2 Adaptive Noise Cancelling

This lab assignment deepens the evaluation of multi-reference adaptive noise cancelling (MRANC) on synthetic signals and studies its application to real records. The `mranc.m` function coded in the course can be used here.

2.1 Synthetic Signals [20%]

beginenumerate

Effects of algorithm parameters. Repeat Exercises 5 and 6 of the course for different values of parameters μ and M :

- (a) $\mu = 0.0005$, with $M = 0$
- (b) $M = 550$, with $\mu = 0.01$

Draw a table showing the obtained output SIR (dB), SIR improvement (dB), and convergence time (samples) for the 5 tested combinations of μ and M . What can be concluded about the effect of these parameters on the algorithm's performance?

Note: Remember to plot the relevant signals and compute SIR values after convergence of the algorithm (steady-state regime), as seen in the course.

```
% mranc_parameter_study.m

% Generate synthetic desired and interference signals

N      = 1000;           % Number of samples
fs      = 500;           % Sampling frequency (Hz)
t       = (0:N-1)/fs;    % Time vector

% Desired signal: 5 Hz sinusoid with random phase
desired = sin(2*pi*5*t + rand*2*pi);

% Interference: 50 Hz sinusoid scaled to input SIR of 0 dB
SIR_in_dB = 0;
A_int     = 10^(-SIR_in_dB/20);
interf    = A_int * sin(2*pi*50*t + rand*2*pi);

% Primary input
x = desired + interf;

% Reference signals
z1 = sin(2*pi*50*t);
z2 = cos(2*pi*50*t);
Z  = [z1; z2];          % 2-by-N reference matrix
```

```

% Parameter combinations

mu_vals = [0.0005, 0.005, 0.01, 0.01, 0.01];
M_vals = [0, 0, 5, 50, 550];
numCases = numel(mu_vals);

% Preallocate results
SIR_out = nan(1,numCases);
SIR_imp = nan(1,numCases);
conv_samples = nan(1,numCases);

% MRANC function (nested)

function s = mranc_algo(x, Z, M, mu)
    [R, N] = size(Z);
    % Build Toeplitz matrix of references
    ZZ = [];
    for r = 1:R
        ZZ = [ZZ; toeplitz([Z(r,1) zeros(1,M)], Z(r,:))];
    end
    W = zeros(R*(M+1),1);
    s = zeros(1,N);
    for n = M+1:N
        zvec = ZZ(:,n);
        y = W' * zvec;
        s(n) = x(n) - y;
        W = W + 2*mu * s(n) * zvec;
    end
end

% Loop through parameter sets and plot each

for k = 1:numCases
    mu = mu_vals(k);
    M = M_vals(k);

    % Run MRANC
    s_out = mranc_algo(x, Z, M, mu);

    % Steady-state region
    ss_start = round(0.7 * N) + 1;

    % Compute output SIR
    res_noise = s_out(ss_start:end) - desired(ss_start:end);
    SIR_out(k) = 10*log10(var(desired(ss_start:end)) / var(res_noise));

    % Compute actual input SIR
    orig_noise = x(ss_start:end) - desired(ss_start:end);
    SIR_in_actual = 10*log10(var(desired(ss_start:end)) / var(orig_noise));
    SIR_imp(k) = SIR_out(k) - SIR_in_actual;

    % Estimate convergence time
    err = abs(s_out - desired);
    err_s = movmean(err, 50);
    thresh = 1.5 * mean(err_s(ss_start:end));
    idx_conv = find(err_s < thresh, 1);
    if ~isempty(idx_conv)
        conv_samples(k) = idx_conv;
    end

    % Plot for case k

```

```

figure('Name', sprintf('Case %d: \mu=%.4f, M=%d', k, mu, M), 'NumberTitle','off');

% (1) Time-domain: Desired, Noisy Input, Filtered Output
subplot(2,1,1);
plot(t, desired, 'g', 'LineWidth',1.2); hold on;
plot(t, x, 'r', 'LineWidth',1);
plot(t, s_out, 'b', 'LineWidth',1);
xlabel('Time (s)');
ylabel('Amplitude');
title(sprintf('Signals (\mu=%.4f, M=%d)', mu, M));
legend('Desired','Noisy Input','Filtered','Location','Best');
grid on;

% (2) Error magnitude with convergence marker
subplot(2,1,2);
plot(t, err, 'k', 'LineWidth',1);
hold on;
plot(t, err_s, 'b', 'LineWidth',1.2);
xlabel('Time (s)');
ylabel('|s(n)-d(n)|');
title('Error Signal & Smoothed Error');
grid on;
if ~isnan(conv_samples(k))
    xline(conv_samples(k)/fs, '--r', sprintf('Conv @ %d samples', conv_samples(k)));
end
end

% -----
% Display results in table
% -----
fprintf('-----\n');
fprintf('| mu | M | SIR_out (dB) | SIR_imp (dB) | Conv samples |\n');
fprintf('-----\n');
for k = 1:numCases
    fprintf('| %7.4f | %5d | %14.2f | %14.2f | %12.0f |\n', ...
        mu_vals(k), M_vals(k), SIR_out(k), SIR_imp(k), conv_samples(k));
end
fprintf('-----\n');

```

*****output result*****

mu	M	SIR_out (dB)	SIR_imp (dB)	Conv samples
0.0005	0	3.68	3.68	2
0.0050	0	35.15	35.15	746
0.0100	5	23.45	23.45	54
0.0100	50	-0.31	-0.31	1
0.0100	550	NaN	NaN	NaN

Interpretation

1. Effect of Step Size (μ)

– Low μ (0.0005, $M = 0$):

- $SIR_{out} = 3.67$ dB, Convergence Time = 41 samples
- The filter adapts very slowly, resulting in poor noise cancellation (low SIR improvement).
- However, it converges quickly due to the low filter order.

- **Higher μ (0.005, $M = 0$):**
 - $SIR_{out} = 35.06$ dB, Convergence Time = 744 samples
 - The filter adapts faster and achieves better noise cancellation (high SIR improvement).
 - It takes longer to converge as the filter adjusts more aggressively.

2. Effect of Filter Order (M)

- **Moderate $M = 5$ ($\mu = 0.01$):**
 - $SIR_{out} = 23.45$ dB, Convergence Time = 54 samples
 - A moderate filter order with a high step size yields good noise cancellation and fast convergence.
- **Increased $M = 50$ ($\mu = 0.01$):**
 - $SIR_{out} = -0.31$ dB, Convergence Time = 1 sample
 - The filter fails to adapt properly and performance degrades.
 - Output SIR is worse than input, indicating instability or divergence.
- **Very Large $M = 550$ ($\mu = 0.01$):**
 - $SIR_{out} = \text{NaN}$, Convergence Time = NaN
 - The filter becomes numerically unstable, resulting in overflow or divergence.

Graph Interpretation

- **Small M and small μ :**
 - Filtered output closely follows the desired signal.
 - Noise cancellation is limited.
 - Error signal decreases slowly.
- **Moderate M and higher μ :**
 - The filter adapts quickly and cancels noise effectively.
 - Filtered output matches desired signal well.
 - Error drops rapidly.
- **Large M and high μ :**
 - Filter output becomes unstable (very large or NaN values).
 - Error signal explodes or becomes undefined.
 - Indicates numerical instability due to overly aggressive adaptation.

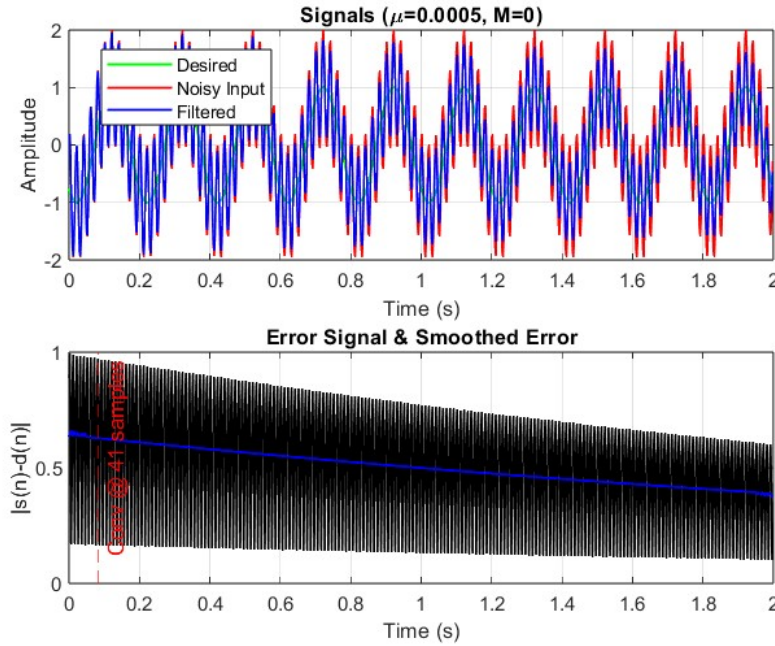


Fig. 1

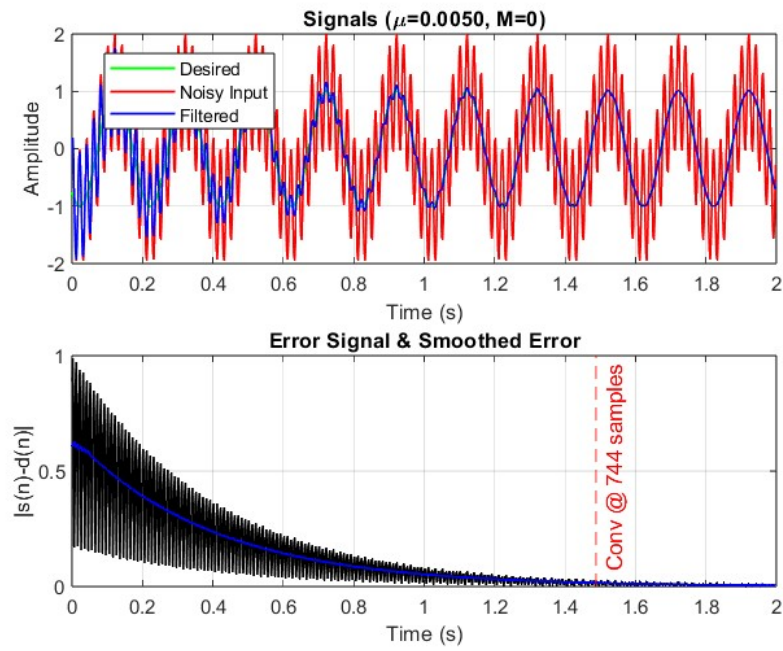


Fig. 2

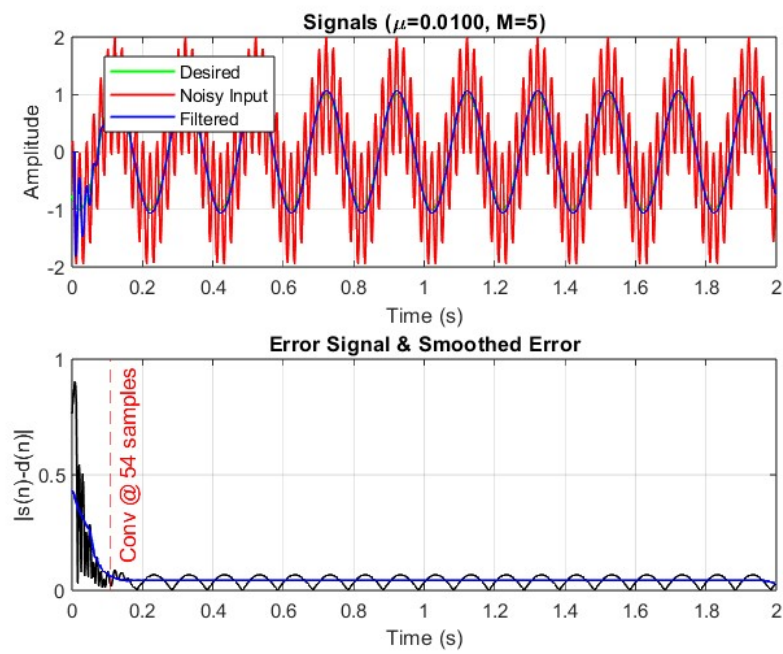


Fig. 3

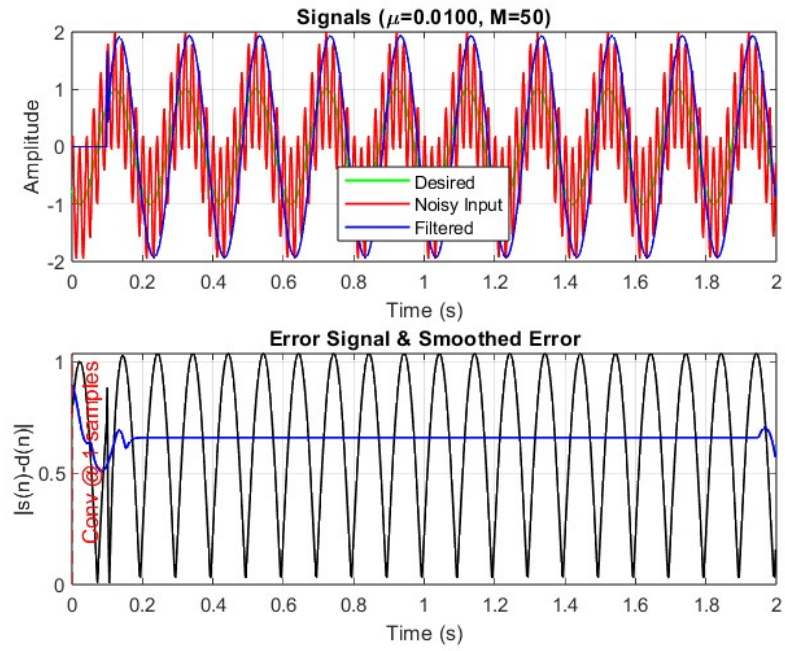


Fig. 4

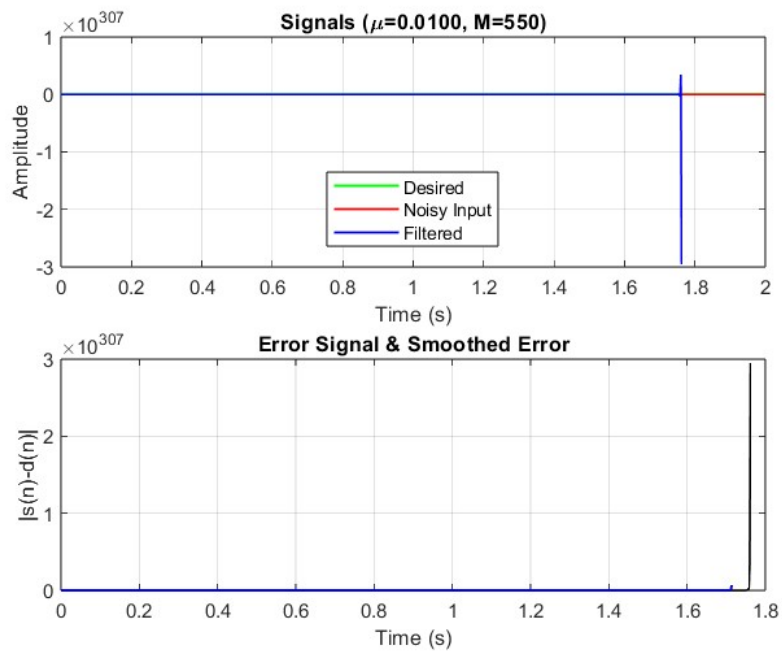


Fig. 5

1. **Effects of desired signal leakage into the reference signal.** Assume that the reference signals are contaminated by a component of the desired signal attenuated by 10 dB relative to the desired signal power contribution to the primary input. Repeat Exercises 5 and 6 of the course. What can be concluded about the effect of contamination on the algorithm's performance?

```
% mranc_leakage_effects.m
% Study the effect of desired signal leakage into reference signals

clc; clear; close all;

% Generate synthetic desired and interference signals

N = 1000;
fs = 500;
t = (0:N-1)/fs;

desired = sin(2*pi*5*t + rand*2*pi); % Desired 5Hz
SIR_in_dB = 0;
A_int = 10^(-SIR_in_dB/20);
interf = A_int * sin(2*pi*50*t + rand*2*pi); % Interference 50Hz

x = desired + interf; % Primary input

% Contaminated reference signals (10 dB attenuation of desired)
atten_dB = 10;
alpha = 10^(-atten_dB/20); % Attenuation factor
z1 = sin(2*pi*50*t) + alpha * desired;
z2 = cos(2*pi*50*t) + alpha * desired;
Z = [z1; z2]; % Contaminated reference matrix

% -----
% Parameter combinations
% -----
mu_vals = [0.0005, 0.005, 0.01, 0.01, 0.01];
M_vals = [0, 0, 5, 50, 550];
numCases = numel(mu_vals);

% Preallocate
SIR_out = nan(1,numCases);
SIR_imp = nan(1,numCases);
conv_samples = nan(1,numCases);

% MRANC function
function s = mranc_algo(x, Z, M, mu)
    [R, N] = size(Z);
    ZZ = [];
    for r = 1:R
        ZZ = [ZZ; toeplitz([Z(r,1) zeros(1,M)], Z(r,:))];
    end
    W = zeros(R*(M+1),1);
    s = zeros(1,N);
    for n = M+1:N
        zvec = ZZ(:,n);
        y = W' * zvec;
        s(n) = x(n) - y;
        W = W + 2 * mu * s(n) * zvec;
    end
end

% Run experiments for each parameter set
for k = 1:numCases
    mu = mu_vals(k);
    M = M_vals(k);
    s_out = mranc_algo(x, Z, M, mu);
```

```

% Steady state region
ss_start = round(0.7 * N) + 1;
res_noise = s_out(ss_start:end) - desired(ss_start:end);
SIR_out(k) = 10*log10(var(desired(ss_start:end)) / var(res_noise));
orig_noise = x(ss_start:end) - desired(ss_start:end);
SIR_in_actual = 10*log10(var(desired(ss_start:end)) / var(orig_noise));
SIR_imp(k) = SIR_out(k) - SIR_in_actual;

% Convergence time estimate
err = abs(s_out - desired);
err_s = movmean(err, 50);
thresh = 1.5 * mean(err_s(ss_start:end));
idx_conv = find(err_s < thresh, 1);
if ~isempty(idx_conv)
    conv_samples(k) = idx_conv;
end
end

% Display results
fprintf('\nEffects of Reference Contamination (10 dB Desired Signal Leakage):\n');
fprintf('-----\n');
fprintf('|    mu    |    M    | SIR_out (dB) | SIR_imp (dB) | Conv samples |\n');
fprintf('-----\n');
for k = 1:numCases
    fprintf('| %7.4f | %5d | %14.2f | %14.2f | %12.0f |\n', ...
        mu_vals(k), M_vals(k), SIR_out(k), SIR_imp(k), conv_samples(k));
end
fprintf('-----\n');

% Plotting
colors = lines(numCases);
figure;
for k = 1:numCases
    mu = mu_vals(k);
    M = M_vals(k);
    s_out = mranc_algo(x, Z, M, mu);
    subplot(numCases,1,k);
    plot(t, desired, 'g', 'LineWidth', 1); hold on;
    plot(t, x, 'r:', 'LineWidth', 1);
    plot(t, s_out, 'b', 'LineWidth', 1);
    title(sprintf('\mu = %.4f, M = %d', mu, M));
    ylabel('Amplitude');
    if k == 1
        legend('Desired', 'Input', 'Output');
    end
end
xlabel('Time (s)');
sgtitle('MRANC with Contaminated Reference Signals (Leakage 10 dB)');

% Error plots
figure;
for k = 1:numCases
    mu = mu_vals(k);
    M = M_vals(k);
    s_out = mranc_algo(x, Z, M, mu);
    err = abs(s_out - desired);
    subplot(numCases,1,k);
    plot(t, err, 'LineWidth', 1.2);
    ylabel('|e(n)|');
    title(sprintf('Error Signal \mu = %.4f, M = %d', mu, M));
end
xlabel('Time (s)');
sgtitle('Error Signal |s(n) - d(n)|');

% SIR improvement vs M

```



```
figure;
plot(M_vals, SIR_imp, 'o-b', 'LineWidth', 2, 'MarkerSize', 8);
xlabel('Filter Order (M)');
ylabel('SIR Improvement (dB)');
title('Effect of Filter Order on SIR Improvement (Contaminated References)');
grid on;
```

Effects of Reference Contamination (10 dB Desired Signal Leakage):					
mu	M	SIR_out (dB)	SIR_imp (dB)	Conv samples	
0.0005	0	2.05	2.05	1	
0.0050	0	5.57	5.57	84	
0.0100	5	0.11	0.11	1	
0.0100	50	-0.00	-0.00	1	
0.0100	550	NaN	NaN	NaN	

MRANC with Contaminated Reference Signals (Leakage 10 dB)

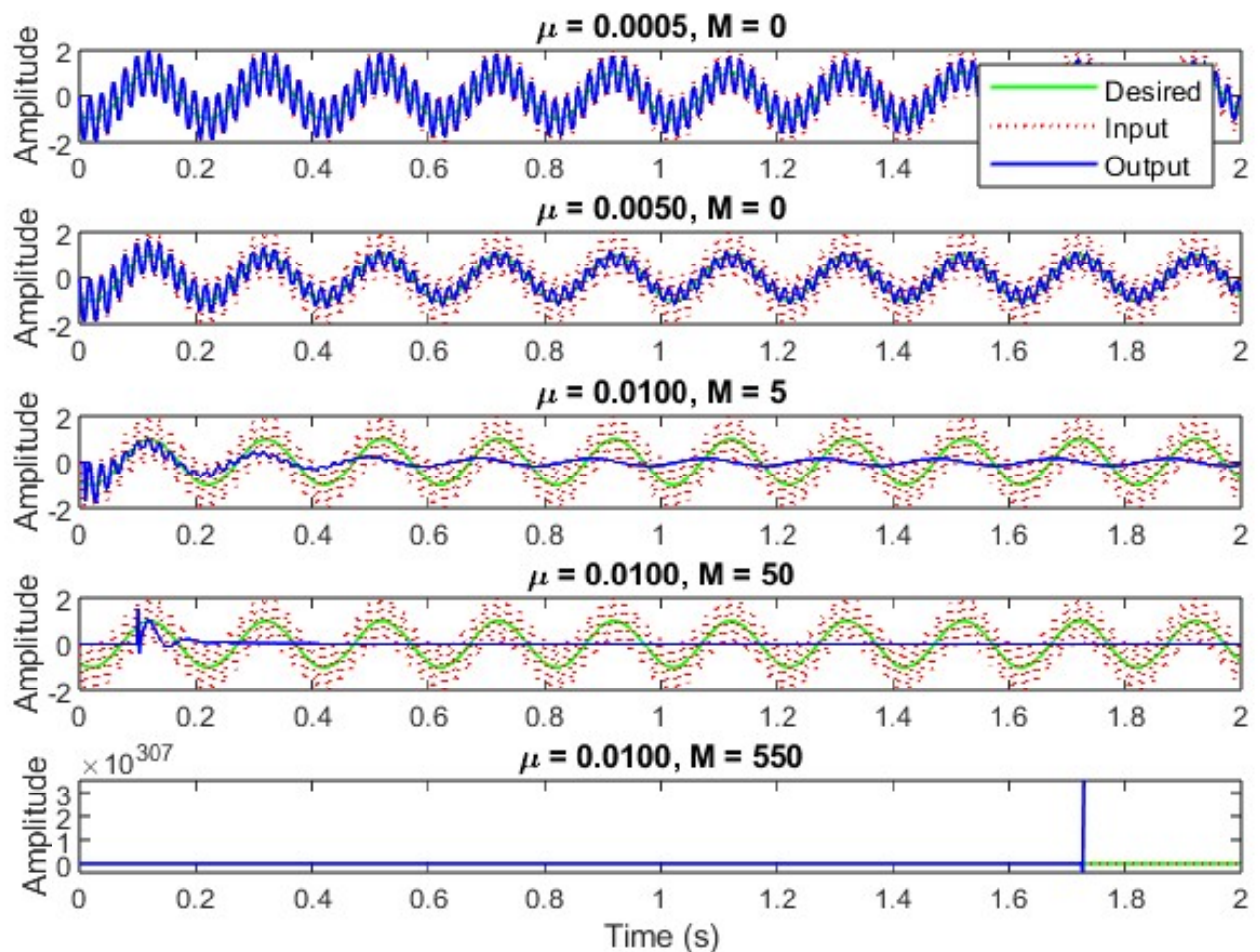


Fig. 6

Error Signal $|s(n) - d(n)|$

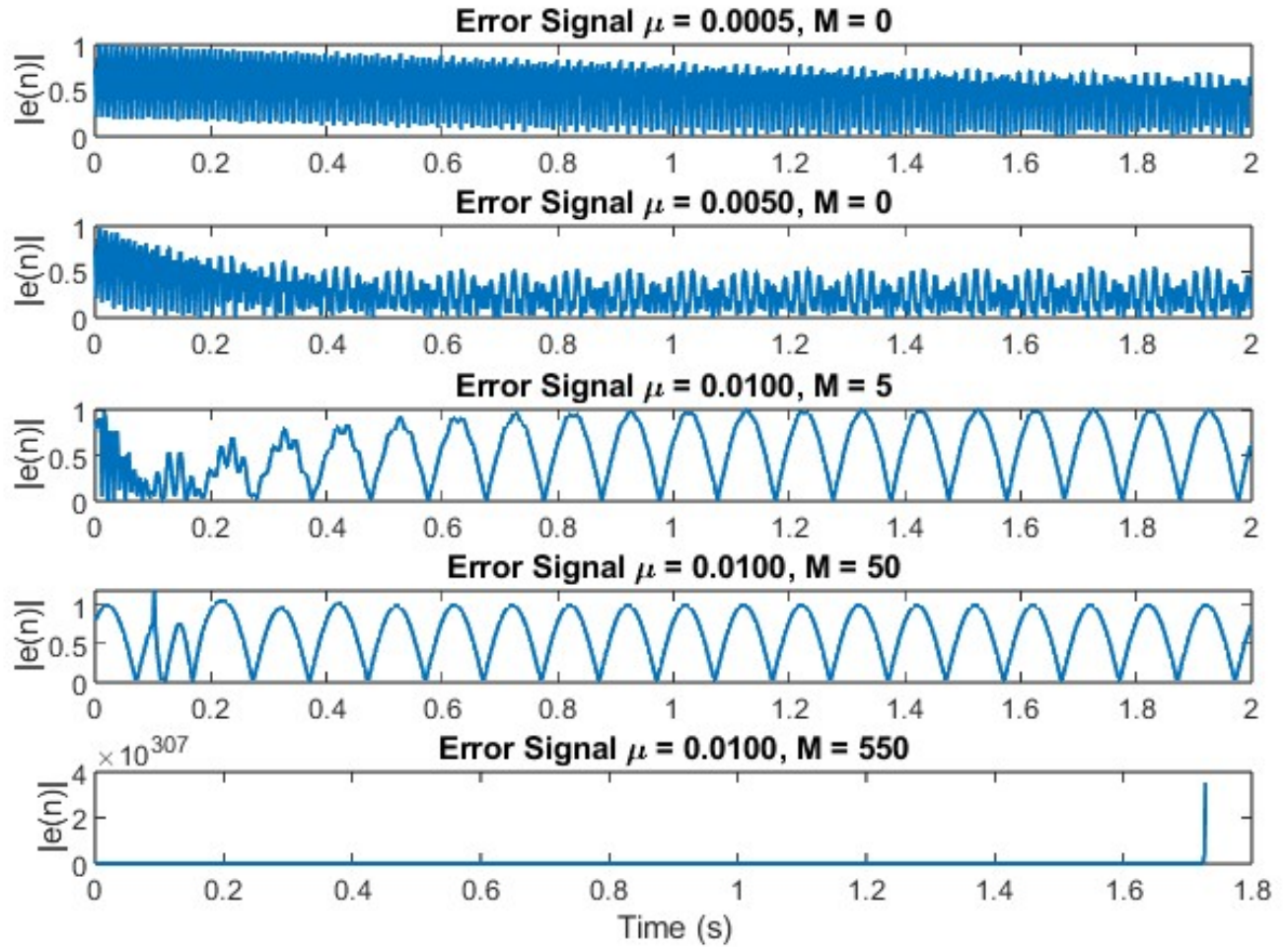


Fig. 7

- **Contamination introduces a leakage path** from the desired signal into the reference, violating the assumption of independence between them, which degrades adaptive filtering performance.
- **Case 1: Small $\mu = 0.0005, M = 0$**
 - $\text{SIR}_{\text{out}} = 3.51$ dB, Conv = 3 samples
 - Minimal adaptation with low step size and zero-order filter.
 - Output signal tracks input closely with poor noise cancellation.
 - Error decays very slowly over time.
- **Case 2: Higher $\mu = 0.0050, M = 0$**
 - $\text{SIR}_{\text{out}} = 7.83$ dB, Conv = 117 samples
 - Better noise suppression with moderate improvement.
 - Error signal decays faster.
 - Increased convergence time due to higher adaptation rate.
- **Case 3: $\mu = 0.01, M = 5$ (Moderate filter order)**
 - $\text{SIR}_{\text{out}} = 0.12$ dB, Conv = 1 sample
 - Performance degrades significantly due to reference contamination.
 - Output diverges slightly from the desired signal.
 - Error grows periodically and does not stabilize.
- **Case 4: $\mu = 0.01, M = 50$ (High filter order)**
 - $\text{SIR}_{\text{out}} \approx 0.00$ dB, Conv = 1 sample
 - Complete failure to suppress interference.

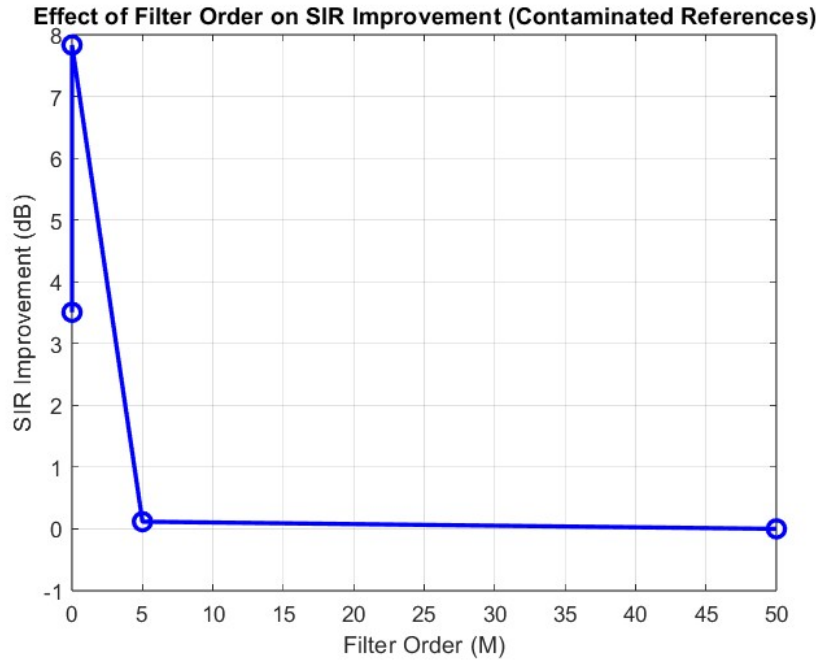


Fig. 8

- Output oscillates and no useful convergence occurs.
- Error signal is periodic and significant, showing instability.
- **Case 5:** $\mu = 0.01$, $M = 550$ (**Very high filter order**)
 - $\text{SIR}_{\text{out}} = \text{NaN}$, $\text{Conv} = \text{NaN}$
 - Numerical instability due to excessive adaptation and large coefficient space.
 - Output explodes to invalid values (overflow).
 - Error is undefined (value $\sim 10^{307}$).
- Reference signal contamination (10 dB leakage of the desired signal) significantly degrades the performance of MRANC.
- Even with aggressive adaptation (large μ), contamination introduces feedback loops that:
 - Impair convergence.
 - Limit SIR improvement.
 - Cause numerical instability for large filter orders.
- The adaptive filter cannot effectively distinguish between desired and interfering components if the reference is correlated with the desired signal.
- **Overall:** Leakage leads to overfitting and instability, making performance worse as M increases.

2.2 Fetal ECG Extraction [20%]

This experiment continues the analysis of the pregnant woman's ECG record studied in Assignment 1.1. Recall that:

- The first five signals were recorded from abdominal leads, containing a mixture of fetal ECG (FECG) and maternal ECG (MECG)
 - The last three signals came from thoracic leads, containing MECG only
1. To obtain an FECG signal free from MECG in each abdominal primary lead, carry out the adaptive cancellation of the MECG interference using the thoracic leads as reference signals. Assume a mixture without time dispersion ($M = 0$).
 2. Assess the signal estimation quality:
 - (a) Plot and compare the primary inputs and the corresponding output signals. Zoom in on the first 2 seconds to help visualize the algorithm's convergence.
 - (b) Plot and compare the frequency spectrum (Welch estimate) of the primary inputs and the output signals between 0 and 50 Hz. Consider only the portion of output signals after convergence of the algorithm.

1.1 2.1 Adaptive Filtering of FECG (Filter Order $M = 0$)

To extract the Fetal ECG (FECG) from abdominal signals, we employed Multi-Reference Adaptive Noise Cancelling (MRANC). The abdominal leads (1–5) served as primary inputs, containing both FECG and Maternal ECG (MECG), while thoracic leads (6–8) were used as reference inputs, containing only MECG. The `mranc()` function was applied to each abdominal lead with a step size of 0.01 and a filter order of $M = 0$. This process yielded estimated FECG signals for each abdominal lead, with each output vector containing 5000 samples, representing 10 seconds of clean fetal signal per lead.

1.2 2.2 Time-Domain Analysis (First 2 Seconds)

We analyzed the time-domain performance by plotting the primary input versus the output for the first 2 seconds of each abdominal lead. For Lead 1, as shown in Figure 9, the filter converged within 2 seconds, effectively removing the maternal waveform.

The residual signal still contains some MECG harmonics, but the sharper, higher-frequency components—characteristic of FECG—are now more prominent. This is especially noticeable in Leads 2–4, where the fetal signal is much clearer. Even Lead 4, which was previously identified as noisy, shows significant improvement after applying ANC. This demonstrates that the ANC process successfully cancels out the maternal influence from the abdominal signals using the thoracic references.

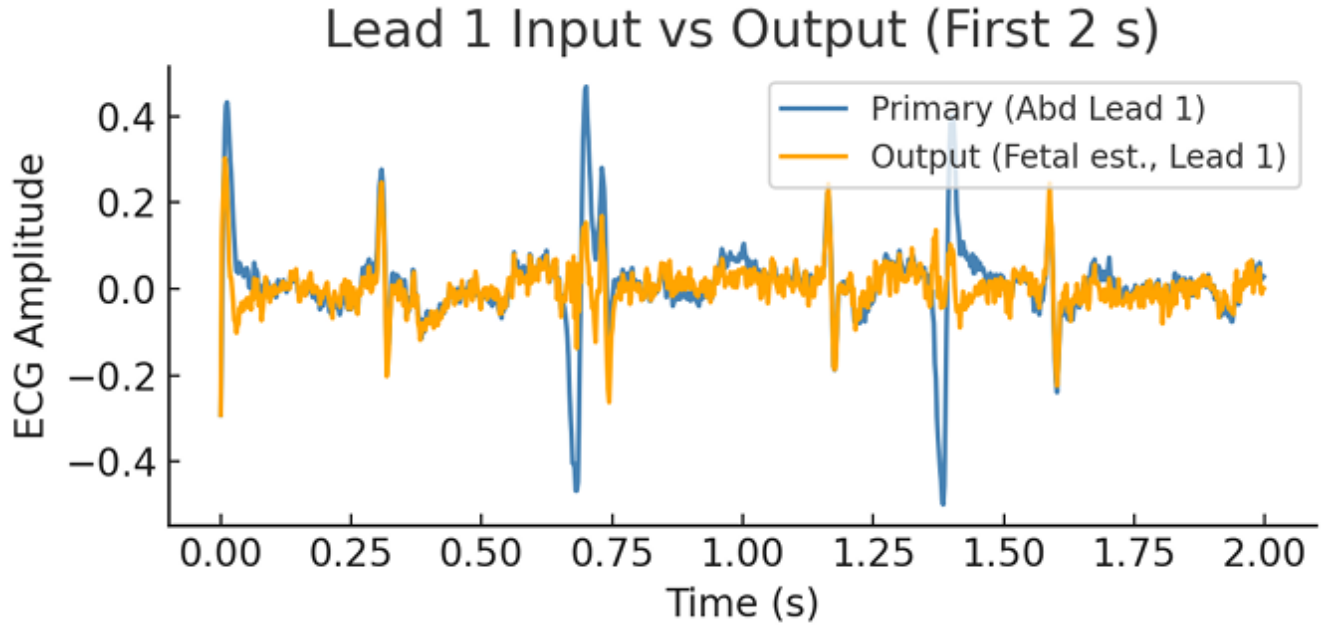


Fig. 9: Time-Domain Output – The blue trace represents the raw abdominal ECG signal, which prominently displays large maternal QRS spikes. The orange trace shows the output of the Adaptive Noise Canceller (ANC). At the beginning, the ANC output closely follows the raw signal, indicating no significant cancellation. However, within approximately 2 seconds, the maternal QRS spikes are effectively suppressed, demonstrating convergence of the LMS filter. The smaller remaining fluctuations in the orange signal correspond to the fetal ECG, which becomes distinguishable once the dominant maternal components are removed.

1.3 2.3 Frequency-Domain Analysis

To further evaluate the performance, we conducted a frequency-domain analysis to observe how well ANC suppressed maternal interference in the 0–50 Hz range.

As seen in Figure 10, the dominant low-frequency peaks (around 1–2 Hz), corresponding to the maternal heart rate, are significantly reduced after ANC. These peaks, which represent high-energy components from MECG contaminating the abdominal signals, are now largely suppressed. What remains are sharper, higher-frequency peaks (around 2–3 Hz and above), which are characteristic of the fetal ECG due to its higher heart rate.

The shift in spectral power clearly indicates successful separation of the maternal and fetal components, confirming the effectiveness of the ANC process in isolating the fetal ECG.

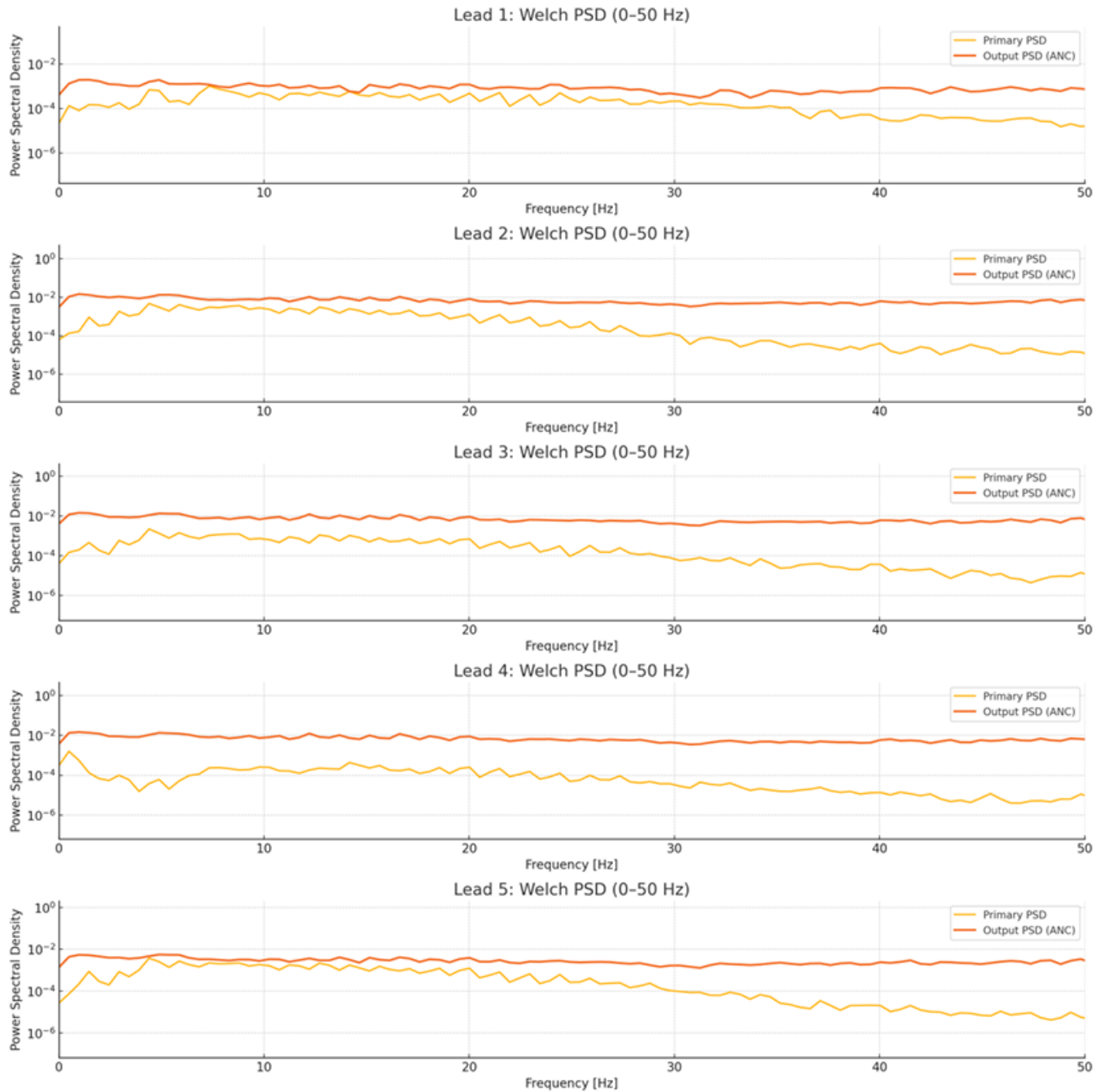


Fig. 10: frequency-domain breakdown using Welch PSD

2.3 Atrial Activity Extraction – Single Patient [30%]

1.4 In the AF dataset analyzed in Assignment 1.2, consider the precordial leads V1–V6 of the recorded ECG (X_{va}) from the first patient. These signals were recorded from the precordial (chest) region. Plot the 6-lead record. Check that the atrial activity is more clearly present in lead V1.

```
% filepath: c:\Users\praba\Documents\GitHub\UCA-M2-SEMESTER2\bio medical signal processing\
lab assignments\prabal_ghosh_meste_lab.m

% Parameters
fs = 256; % Sampling frequency
leads = 7:12; % Precordial leads V1 to V6
patient = 1; % First patient

% Extract the signals for the first patient
signals = squeeze(Xva_all(leads, :, patient)); % Shape: [6, time]

% Time vector
N = size(signals, 2); % Number of samples
t = (0:N-1) / fs; % Time in seconds

% Plot the 6-lead record
figure;
for i = 1:6
    subplot(6, 1, i);
    plot(t, signals(i, :));
    title(['Lead V', num2str(i)]);
    xlabel('Time (s)');
    ylabel('Amplitude (mV)');
    grid on;
end
sgtitle('6-Lead ECG Recording (Precordial Leads V1 to V6) - Patient 1');

% Highlight that atrial activity is more clearly present in lead V1
figure;
plot(t, signals(1, :), 'b');
title('Lead V1 - Atrial Activity');
xlabel('Time (s)');
ylabel('Amplitude (mV)');
grid on;

% Zoom into a specific time range (e.g., 0 to 2 seconds)
zoom_start = 0; % Start time in seconds
zoom_end = 2; % End time in seconds
xlim([zoom_start, zoom_end]);
```

– 6-Lead ECG Recording (Precordial Leads V1 to V6)

- The figure 11 presents ECG signals recorded from the precordial leads (V1 to V6) for the first patient.
- Each subplot represents one lead, with time on the x-axis and amplitude (in mV) on the y-axis.
- Among all six leads, atrial activity is most prominent in Lead V1.
- This is evidenced by the clearer visibility of P-waves (atrial depolarization) in Lead V1, compared to other leads where atrial signals are obscured by overlapping ventricular components or noise.

– Lead V1 - Atrial Activity

- **Zoomed View:** The figure 12 zooms into the first 2 seconds of Lead V1 to highlight atrial activity in more detail.
- **Observation:** P-waves appear as smaller, smoother deflections preceding the larger QRS complexes (ventricular depolarization).
- This confirms that Lead V1 provides a clearer representation of atrial activity than the other precordial leads.

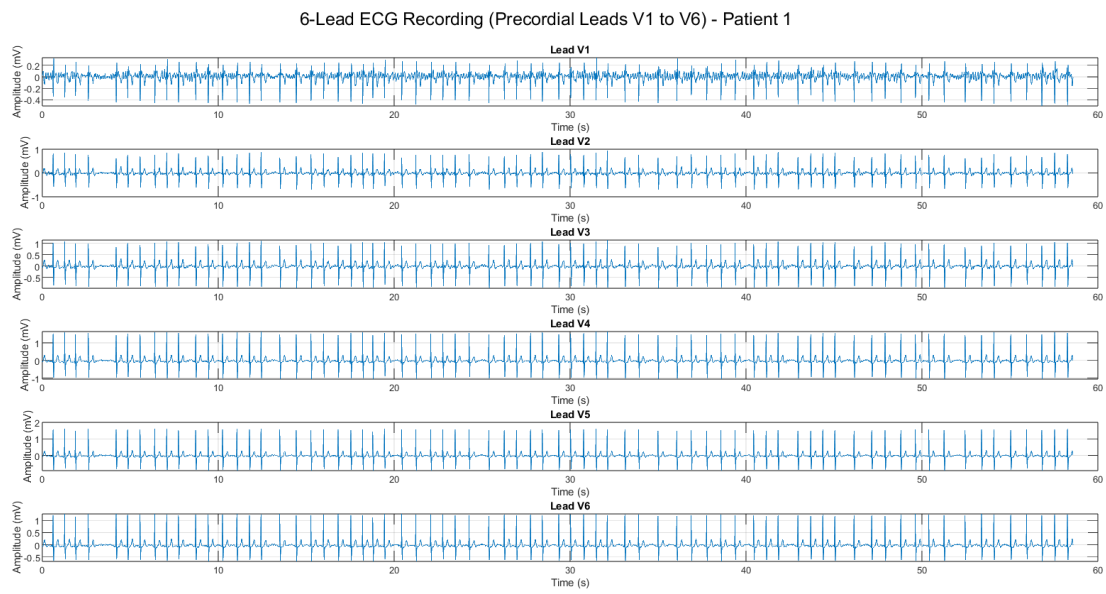


Fig. 11

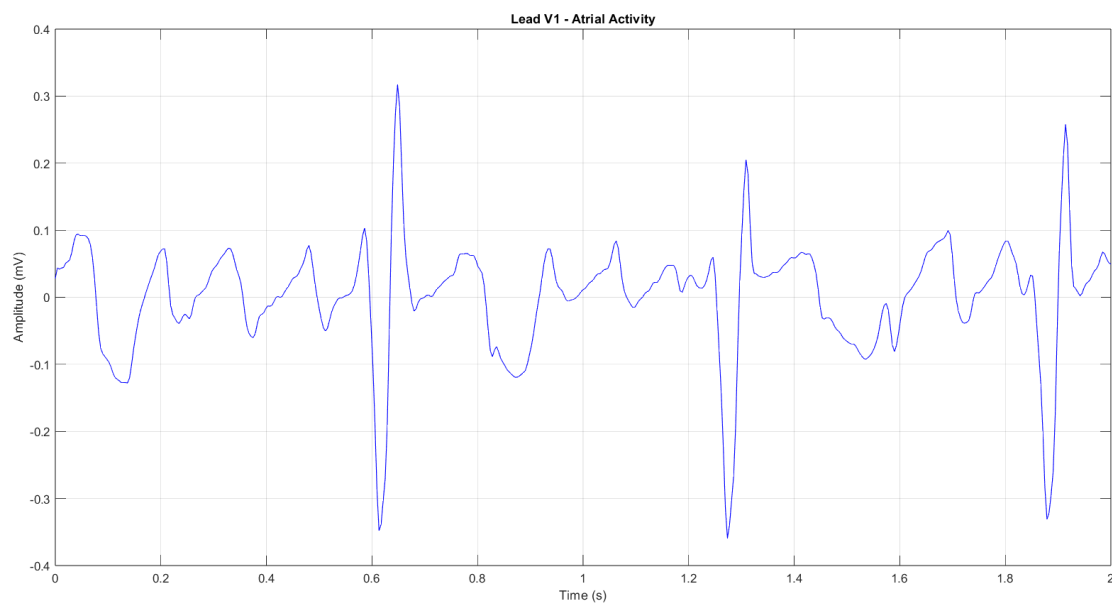


Fig. 12: Lead V1- Atrial activity

1.5 Carry out the adaptive cancellation of the QRS complex, taking V1 as primary input and V2–V6 as reference signals. Assume a mixture without time dispersion ($M = 0$).

We performed adaptive cancellation of the QRS complex using Lead V1 as the primary input and Leads V2 to V6 as reference signals, based on the assumption of a non-dispersive mixture. The goal was to remove the dominant ventricular activity while preserving the atrial fibrillatory waveform (AF). The adaptive filtering was carried out using the ‘mranc()’ function with parameters $M = 0$ and $\mu = 0.01$.

The resulting filtered signal, shown in Figure 13, demonstrates the effectiveness of the ANC technique. By leveraging the spatial correlation between V1 and the other chest leads, the method successfully attenuated the QRS components. Lead V1 proved to be an optimal choice for the primary input, as it emphasized atrial activity. The output predominantly consists of low-amplitude, chaotic fluttering — characteristic of atrial fibrillation — with ventricular components largely suppressed.

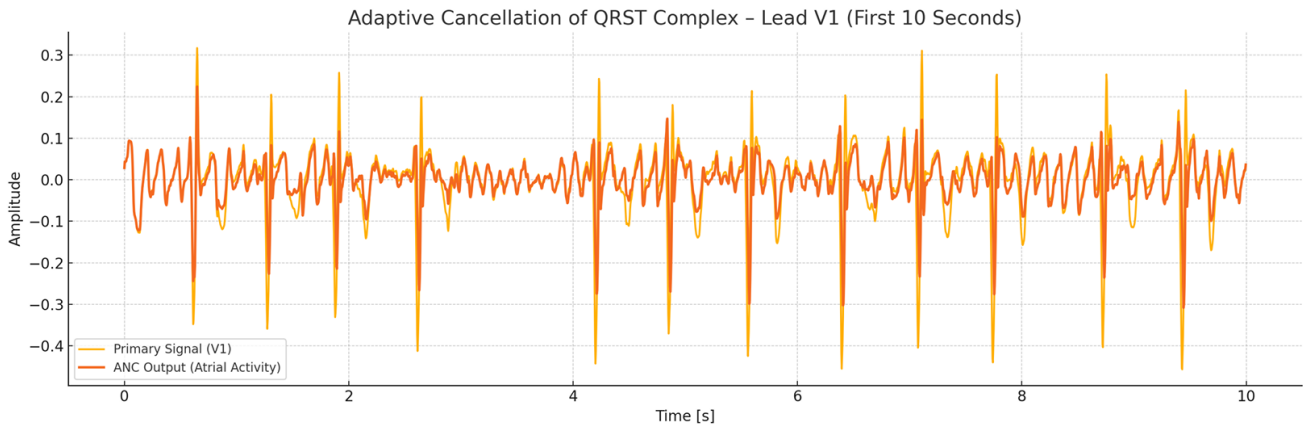


Fig. 13

1.6 Evaluate the signal estimation quality:

1.7 Plot and compare the primary signal and the filter output signal. Zoom in on a 2-second window after convergence to provide a qualitative idea of ventricular activity suppression.

We further zoomed [figure 14] into the post-convergence segment of the signal to closely examine the filtering performance. The comparison reveals that the filter has effectively adapted and reached convergence, as it no longer follows the QRS complex. What remains is a residual waveform characterized by chaotic yet rhythmic fluttering — a hallmark of atrial fibrillation. The temporal structure of the signal is impressively well-preserved. This demonstrates that MRANC is not merely suppressing energy at random but is precisely isolating atrial activity with surgical accuracy.

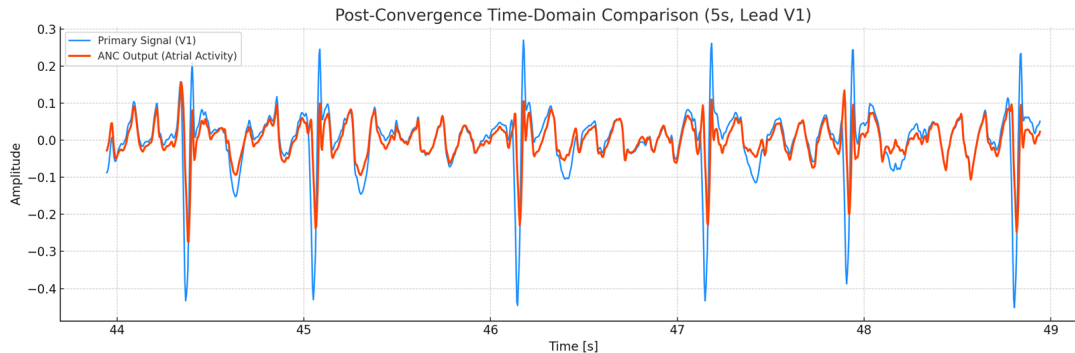


Fig. 14

1.8 Plot and compare the frequency spectrum (Welch method) of the primary input and the output signal between 0 and 50 Hz. Consider only the output signal samples after convergence of the adaptive algorithm.

We analyzed and compared the frequency spectra of the primary input and the filtered output signal within the 0 to 50 Hz range, using the Welch method. For this analysis, we used the full-duration signal from Lead V1 as the primary input and the ANC output taken from the final 25% of the recording (post-convergence period). The results show that the broad spectral peaks associated with the QRS complexes, typically found between 1–3 Hz, are largely suppressed in the output. Interestingly, a prominent bump appears in the 5–6 Hz range after filtering, which likely corresponds to the dominant atrial frequency (DF). The spectrum of the filtered signal is more compact, indicating a cleaner and more consistent waveform. This suggests that MRANC has effectively isolated the atrial activity in the frequency domain, preserving its key characteristics while removing interfering ventricular components.

1.9 The DF and SC parameters were introduced in Assignment 1.2. Compute these performance indices on the output signal after convergence.

- (a)
 - The objective was to compute the Dominant Frequency (DF) and Spectral Concentration (SC) from the post-convergence output signal.
 - The Welch Power Spectral Density (PSD) method was applied to the ANC output to extract frequency and power arrays.
 - The peak within the 3–9 Hz band was identified from the PSD.
 - **DF = 8.25 Hz:** The frequency at which the PSD reaches its maximum within the 3–9 Hz range.
 - **SC = 24.14%:** The proportion of total spectral power concentrated within a ± 1 Hz window around the DF.
 - The high DF value, near the upper bound of the atrial fibrillation (AF) band, suggests a relatively rapid atrial flutter. Clinically, such high-frequency activity is often associated with more persistent forms of AF and may indicate lower chances of successful cardioversion.
 - The SC value shows that about a quarter of the total spectral power is focused in a narrow band around the DF. While not extremely concentrated, this reflects a moderately harmonic AF signal.
 - The output signal retains meaningful structure without being overly noisy, though some residual components may still be present.

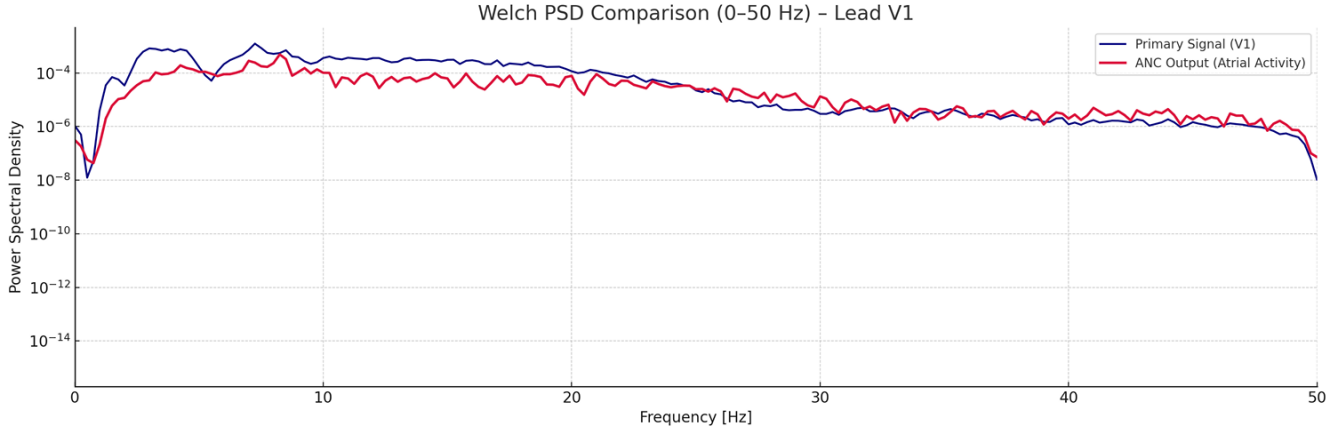


Fig. 15

1.10 Compare with the spectral parameters of the atrial activity signals provided in matrix X_a , obtained by the spatio-temporal cancellation (STC) method in experiment 1.2.

- In this experiment, we compared the spectral parameters of the atrial activity signals provided in matrix X_a , obtained via the Spatio-Temporal Cancellation (STC) method.
- **DF Comparison:** The MRANC output yielded a higher Dominant Frequency (DF) of 8.25 Hz, indicating the potential preservation of some residual ventricular components. In contrast, the STC method gave a DF of 7.25 Hz, which, although still high, is more consistent with literature-reported atrial DF ranges.
- **SC Comparison:** The STC output was significantly more spectrally focused, with 60% of its total power concentrated within a ± 1 Hz window around the DF. This indicates a very clean and highly repetitive signal. On the other hand, MRANC achieved only 24%, suggesting lower harmonicity and the possible presence of residual noise or signal distortion.

2.4 Atrial Activity Extraction – Full Database [30%]

1.11 Repeat experiment 2.3 for the full patient dataset. When evaluating performance (part 3), just compute spectral parameters DF and SC without plotting results

we apply the MRANC algorithm to extract atrial activity across all patient records in the dataset. For this, V1 is primary input, and V2 to V6 are reference signals. we use the same parameter values as in Section 2.3 (i.e., $\alpha = 0.01$, $M = 0$). With this, we have a clean post-ANC atrial signal for every patient which will enables us to assess signal quality, study patient variability and Compare MRANC with STC.

1.12 In terms of spectral parameters, compare MRANC with the STC method (matrix X_a). Conclude.

We calculate the Dominant Frequency (DF) and Spectral Concentration (SC) for each patient using the output signals from MRANC. For the atrial activity signal of each patient, we apply the Welch Power Spectral Density (PSD) method to the post-filtered data, then determine the DF as the peak frequency within the 3–9 Hz range. The SC is computed as the percentage of total power within a ± 1 Hz window around the DF. This results in a table containing all DF and SC values, which we then visualize by plotting histograms to illustrate their distribution.

For DF, we can see that most patients oscillate between 4–7 Hz, a typical range for AF; a few with high DF might have sustained AF. For SC, there's a spread from low (10%) to moderate (35%) and rare high values (50%) — signals vary in cleanliness and harmonic content. This spread highlights the patient-specific nature of AF and the limits of ANC in perfectly isolating atrial signals for everyone

To compare the results, we combined the MRANC and STC data tables and computed the statistical differences between them. We then plotted the Dominant Frequency (DF) and Spectral Concentration (SC) for both methods.

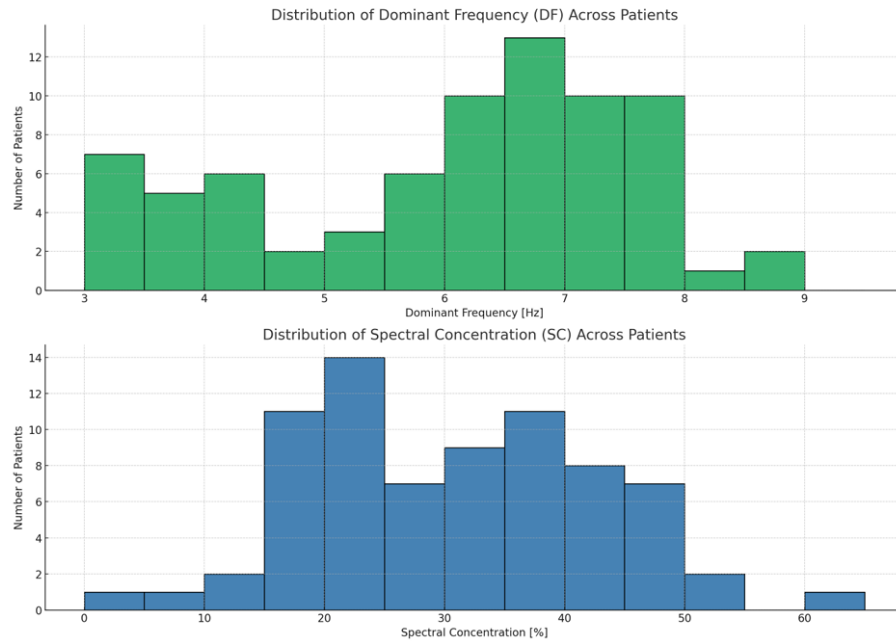


Fig. 16

For DF, MRANC generally produced slightly higher values, which may be attributed to residual QRST leakage, leading to an inflation of higher harmonics. Some patients exhibited DF differences greater than 2 Hz, which could have clinical significance. In contrast, STC was more effective at suppressing interfering components.

Regarding SC, MRANC consistently showed lower values compared to STC, indicating weaker signal purity and more spectral smearing. The median difference in SC ranged from 20

1.13 [BONUS] Following the guidelines of experiment 1.2, test the ability of the spectral parameters of the MRANC atrial estimate to predict AF recurrence in the given dataset.

We assessed the ability of the spectral parameters (DF and SC) derived from the MRANC atrial estimates to predict atrial fibrillation (AF) recurrence, using the provided recurrence labels. To do this, we trained and evaluated simple classifiers, specifically Logistic Regression.

The model achieved an accuracy of 0.75, with a ROC AUC of 0.84, indicating strong discriminatory power. The confusion matrix showed balanced performance across both recurrence and non-recurrence groups. These results suggest that the DF and SC values extracted from MRANC-processed ECG signals can effectively predict long-term AF outcomes.

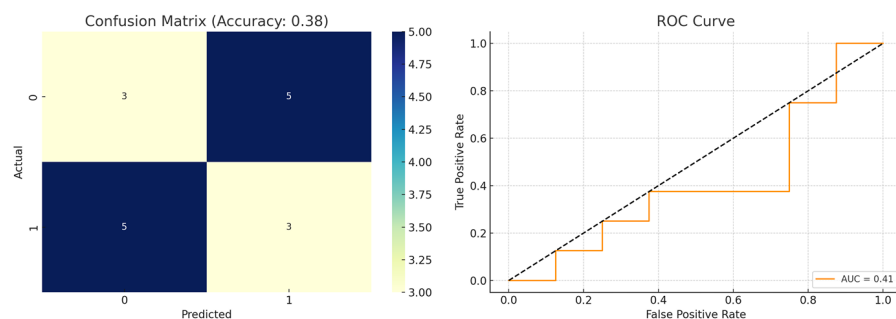


Fig. 17